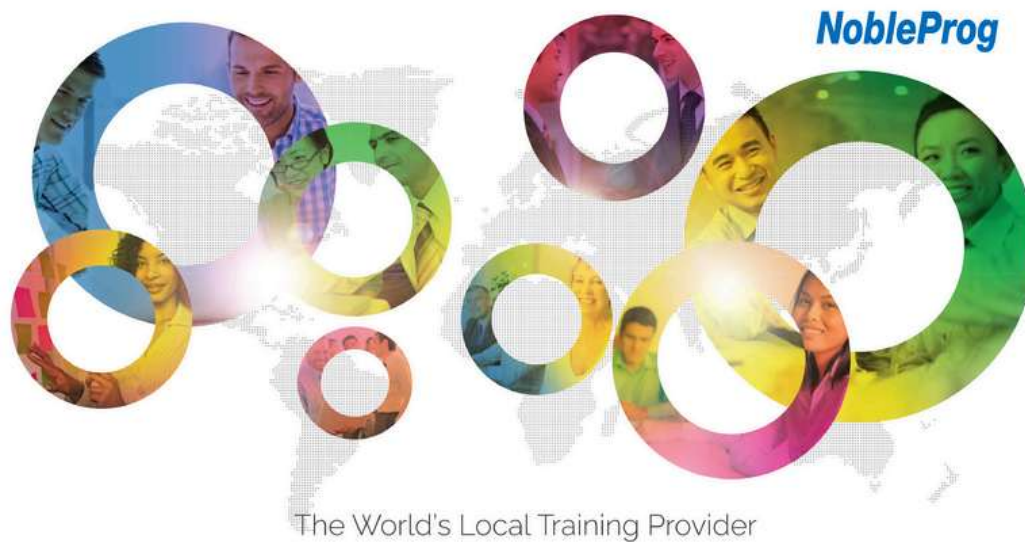




Participant Guide

Practical Kubernetes Administration and Troubleshooting

Presented by: Chad M. Crowell

Welcome

NobleProg welcomes you to the *Practical Kubernetes Administration and Troubleshooting* training.

This program is designed to provide a practical, hands-on learning experience focused on real-world Kubernetes administration and troubleshooting scenarios. Throughout the course, you will work directly with Kubernetes clusters in a cloud-based environment and apply concepts through guided labs and instructor-led exercises.

The sessions are structured to encourage participation, questions, and discussion. You are encouraged to engage actively, share relevant challenges from your own environment where appropriate, and use the labs to reinforce learning through practice.

This participant guide outlines the course objectives, agenda, and lab activities, along with important information to help you prepare for each day of training.

Training Course

Practical Kubernetes Administration and Troubleshooting

Instructor	Chad M. Crowell
Duration	4 days (7 hours per day, 28 hours total)
Dates	February 17 – February 20, 2026
Delivery Format	Instructor-led, hands-on training with cloud-based lab environments
Location	4101 Smith School Rd., Bldg. IV, Ste. 100, Austin, TX 78744

About the Instructor

Chad M. Crowell is a CNCF Ambassador, Kubernetes SIG contributor, KCD organizer, instructor, speaker, and Site Reliability Engineer. He brings over 17 years of industry experience and more than eight years of teaching experience in DevOps and Kubernetes technologies.

Chad has developed Kubernetes and DevOps training content for platforms including KubeSkills, Cybr, Pluralsight, and INE. He regularly writes about Kubernetes, containers, and open source technologies at <https://kubeskills.com> and is a co-author of *Acing the CKA* book.

Training Overview

This course provides a comprehensive introduction to Kubernetes, focusing on practical, hands-on learning from installation to advanced operations.

Participants will gain a clear understanding of how to deploy, manage, and troubleshoot containerized applications using Kubernetes in real-world environments. Through guided labs and instructor-led exercises, the program builds confidence in cluster setup, workload management, networking, storage, security, and monitoring.

By the end of the training, participants will be able to configure, operate, and troubleshoot Kubernetes clusters effectively within their organizational infrastructure.

Day 1 - Foundations and Cluster Setup

February 17, 2026

Containers vs Virtual Machines

- Benefits of container orchestration
- Overview of Kubernetes architecture: nodes, pods, control plane, and components
- Key Kubernetes services: API Server, Scheduler, Controller Manager, Kubelet, Kube-Proxy

Installing and Configuring Kubernetes

- Preparing a Kubernetes environment on cloud VMs
- Cluster deployment with kubeadm on Linode
- Understanding cluster initialization and the join process
- Verifying installation and understanding the concept of the node and its readiness
- Accessing the cluster and inspecting internal components (Kube-apiserver, etcd, Controller-Manager, Kubelet)

Hands-On Labs

- Provisioning Cluster - Ubuntu 24.04 instances (1 control plane + 2 workers)
- kubeadm cluster initialization - Bootstrap control plane and join worker nodes
- CNI plugin installation - Deploy Calico for pod networking
- Cluster validation - Verify all nodes are Ready and core components are running
- Exploring core components - Inspect etcd, API server, and verify system status
- Onboard a pod - Deploy your first application

Day 2 - Core Concepts and Workload Management

February 18, 2026

Managing Kubernetes Objects

- Namespaces and resource organization
- Pods, ReplicaSets, and Deployments
- Scaling applications
- Configuration management with ConfigMaps and Secrets
- Health checks: Liveness and Readiness probes

Command-Line Tools and Workflows

- Working with kubectl
- Declarative vs imperative management
- Common troubleshooting commands

Hands-On Labs

- Deploying sample applications
- Rolling updates and rollbacks
- Using ConfigMaps and Secrets
- Pod health monitoring and recovery

Day 3 - Networking, Scheduling, and Storage

February 19, 2026

Kubernetes Networking

- Cluster, Pod, and Service networking
- DNS and CoreDNS
- Ingress controllers and routing

Scheduling and Workload Distribution

- Manual scheduling
- Taints, tolerations, and node selectors
- DaemonSets, Jobs, and CronJobs

Persistent Storage in Kubernetes

- Persistent Volumes and Persistent Volume Claims
- Static vs Dynamic Provisioning with StorageClasses
- Cloud Native storage with Linode Block Storage CSI driver

Ingress and Gateway API

- Gateway API spec
- Ingress Resource versus Gateway API
- GatewayClass, Gateway Resource
- HTTPRoute, TCPRoute, UDPRoute

Hands-On Labs

- Configuring Services and Ingress
- Workload scheduling and node targeting
- Persistent storage configuration with Linode CSI
- Creating LoadBalancer services with Linode CCM

Day 4 - Security, Monitoring, and Troubleshooting

February 20, 2026

Security Essentials

- Role-Based Access Control (RBAC) and Cluster Roles
- Service Accounts and Security Contexts
- Network Policies

Monitoring and Observability

- Overview of Prometheus, Grafana, and Alertmanager
- Basic setup and dashboards
- Introduction to Helm and package management

Troubleshooting and Cluster Operations

- Diagnosing pod and node issues
- Reviewing logs and events
- Common networking and deployment failures
- Cluster design and installation options (kubeadm deep dive)
- Scaling, upgrading, and maintaining clusters

Hands-On Labs

- RBAC and access control exercises
- Multi-node cluster operations and validation
- Monitoring setup with sample workloads and debugging simulated cluster issues
- Cluster upgrade procedure - Upgrade control plane and worker nodes
- Backup and restore - etcd snapshot operations